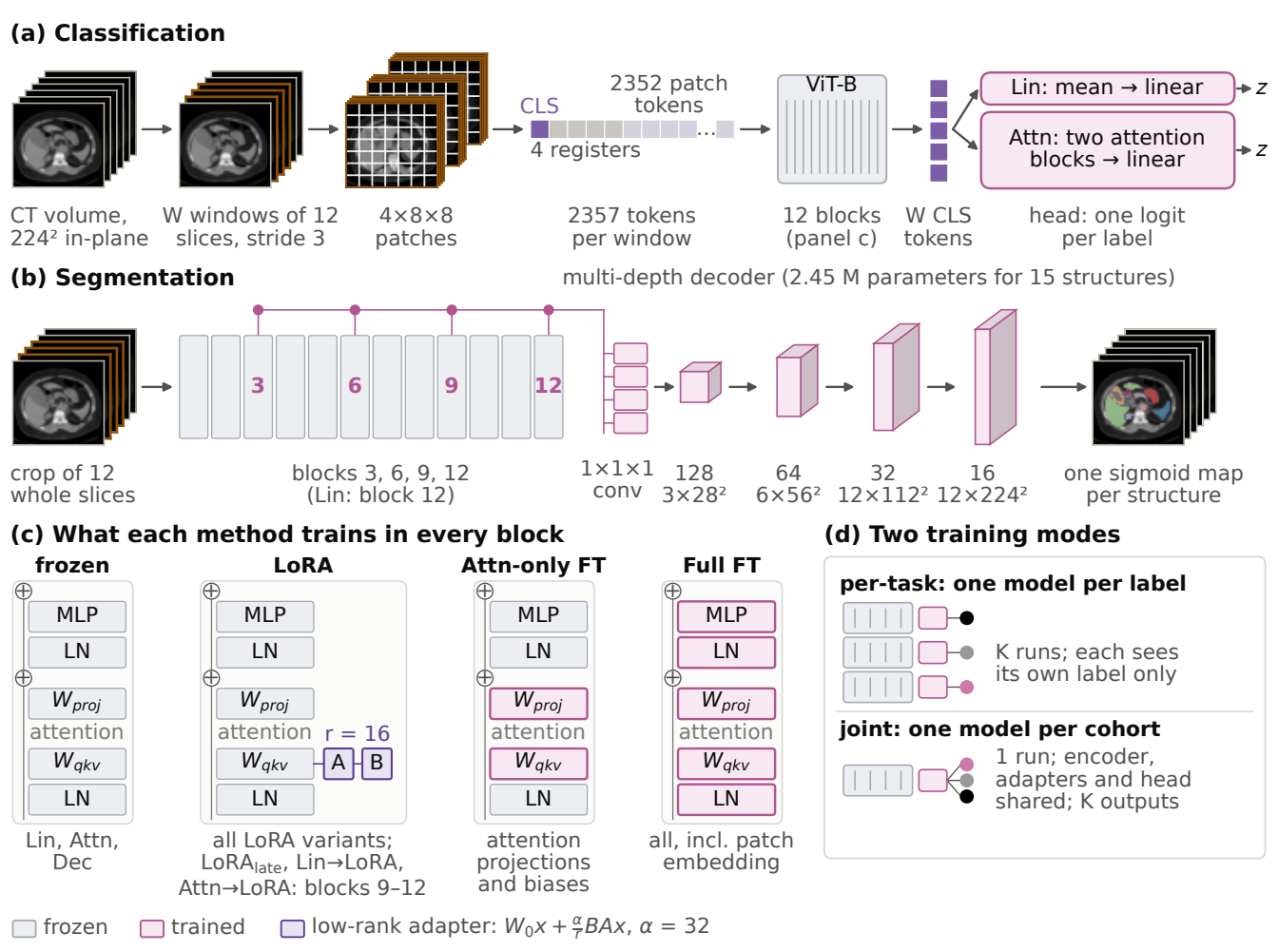

(a) Classification
CLS
2352 patch tokens
4 registers
ViT-B
Lin: mean → linear
Attn: two attention blocks → linear
z
z
CT volume, 224² in-plane
W windows of 12 slices, stride 3
4×8×8 patches
2357 tokens per window
12 blocks (panel c)
W CLS tokens
head: one logit per label
(b) Segmentation
multi-depth decoder (2.45 M parameters for 15 structures)
3
6
9
12
crop of 12 whole slices
blocks 3, 6, 9, 12 (Lin: block 12)
1×1×1 conv
128 3×28²
64 6×56²
32 12×112²
16 12×224²
one sigmoid map per structure
(c) What each method trains in every block
frozen
LoRA
Attn-only FT
Full FT
MLP
LN
W_{proj}
attention
W_{qkv}
LN
r = 16
A
B
Lin, Attn, Dec
all LoRA variants; $LoRA_{late}$, Lin→LoRA, Attn→LoRA: blocks 9–12
attention projections and biases
all, incl. patch embedding
(d) Two training modes
per-task: one model per label
K runs; each sees its own label only
joint: one model per cohort
1 run; encoder, adapters and head shared; K outputs
frozen
trained
low-rank adapter: $W_0x + \frac{\alpha}{r}BAx$, $\alpha = 32$